Applied Soft Computing Journal 171 (2025) 112793

Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Recognition of Egyptian hieroglyphic texts through focused generic segmentation and cross-validation voting

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HIGHLIGHTS

- Largest dataset of Egyptian hieroglyphs, with 310 classes and nearly 14,000 images.
- Hieroglyphs recognition by means of a joint segmentation and classification model.
- Use of focused generic segmentation integrated with SAM to extract hieroglyphs.
- Ensemble model of ConvNeXt backbones using Cross-Validation-Voting (CVV).
- Better classification results of the state of the art, improving detection models

ARTICLE INFO

Keywords:
Egyptian hieroglyphs
Generic segmentation
SAM
CVV
ConvNeXt

ABSTRACT

Ancient Egyptian hieroglyphs form part of a complex language that has attracted the attention of Egyptologists, historians, and amateurs for centuries. In use for more than 3000 years, they consist of hundreds of symbols that can be transcribed into their Latin phonemes. Although there have been some previous works on the recognition of hieroglyphs through computer vision, this is a study of unprecedented depths and presents several unique contributions. On the one hand, we have created the largest and most complete dataset of existing Egyptian hieroglyphs to date, covering all the main symbols used on stelae. On the other, we have carried out a systematic analysis of detection, segmentation, and classification methods, focusing our research on a composite method of focused generic segmentation and classification with an ensemble model of ConvNeXt backbones using Cross-Validation Voting (CVV). Our trained model has been evaluated against several carved or painted stone stelae, obtaining excellent results. To the best of our knowledge, there is currently no other methodology capable of obtaining the classification results presented in this paper, and the method and the dataset presented represent a very significant advancement in the development of automated methods for reading Egyptian hieroglyphic texts.

1. Introduction

Egyptian hieroglyphs can be found in texts from the beginning of Pharaonic times. Ancient Egyptians drew inspiration from the world around them to create the symbols and sigils used for their writing, transforming animals, plants, natural elements, domestic objects, and buildings into semi-abstract symbols. It was around 200 years ago that Jean-François Champollion began to decipher these hieroglyphs, after the discovery of the Rosetta Stone a few years earlier by an officer of Napoleon's Corps of Engineers. Hieroglyphs, as sacred signs, were used to decorate the walls of temples, tombs, stelae, papyri, and other monuments, and inscriptions were ascribed the magical powers that the ancient Egyptians attributed to the word. The texts in which they were incorporated had no fixed writing direction and could be read from left to right or right to left, as well as from top to bottom, depending on whether they were in rows or columns; the human or animal figures indicated the reading direction of the text. In addition, each of the hieroglyphs can have several meanings depending on the context. They can be ideograms (a single hieroglyph represents only a certain idea or thing), determinatives (indicating to which category a word belongs) and phonograms (when a hieroglyph represents one, two, three or several sounds).

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https://doi.org/10.1016/j.asoc.2025.112793
Received 17 March 2024; Received in revised form 26 July 2024; Accepted 13 January 2025
Available online 23 January 2025
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There are previous works in the literature that address the problem of hieroglyphic detection using classical computer vision techniques [1–7]. However, these techniques worked only under very limited conditions and were not able to generalize as well as deep neural networks. Other approaches have attempted to detect a small number of hieroglyphs using detection networks such as YOLO or Detectron2 [8–10]. However, from our own preliminary experiments, these networks do not provide accurate results and require more complex labeling of hieroglyphs, such as including hieroglyphs with various backgrounds with other hieroglyphs. Hieroglyphic labeling is a costly process requiring the involvement of experts with extensive knowledge of this field.

In this paper we propose the idea of using a generic focused SAM-type segmentation model. The Segment Anything Model (SAM) [11] is a generic segmentation network able to work in a zero-shot learning problem. It receives different kinds of prompts and segments, known and unknown objects, and parts of the scene based on these prompts. By focused, we mean a set of vision techniques that force the SAM to act on specific prompts. This allows the SAM to segment hieroglyphs one by one, instead of performing a global segmentation. The result of global SAM segmentation is not usable for our purpose. Once the hieroglyphs are segmented, we perform a process of extracting the hieroglyphic layout and then classifying them one by one using Cross-Validation Voting [12], a technique which, as we will see, greatly improves the classification result. The ensemble CVV model uses ConvNeXt backbones [13]. In addition to the results of the isolated classification, we also show various classifications of stelae made with our model, including phonetic transliteration and recreation in JSesh [14,15].

We also present the largest dataset of Egyptian hieroglyphs ever created, including 13,729 hieroglyphs labeled in 310 classes. We follow the specific nomenclature of Egyptologists and make this dataset and the models available to the scientific community.

The effects of time, exposure and even vandalism make the hieroglyph recognition process even more difficult. In our case, hieroglyphs are also represented in different materials i.e. concretely carved or painted stone stelae. For these reasons, a solution is required that could be utilized in various different situations. We have made an extensive survey of all the works and technologies that have been applied to the problem of hieroglyph recognition, and our method significantly improves the current state of the art, both in the number of hieroglyphs detected, the type of stelae, and the quality of recognition. The main contributions of this research can be summarized as follows:

- A hieroglyph recognition method using focused generic segmentation and ensemble classification models with cross-validation voting (CVV). The segmentation strategy uses the Method Based on Region Segmentation (MBRS), using classic computer vision techniques, backed up by the Integrated Generic Segmentation Model (IGSM), based on the SAM, as an alternative to the former for cases where edges are not clearly defined. The classification method uses the ConvNeXt backbone and the CVV classification technique, which allows the use of different similar classifiers trained on disjoint validation sets.
- The largest existing hieroglyphic classification dataset, with 310 classes and 13,729 labeled images. This dataset is an invaluable resource for anyone studying ancient Egyptian history. It encompasses all periods, from the Old Kingdom to the New Kingdom, and is therefore essential for anyone seeking to gain a comprehensive understanding of the evolution of hieroglyphic writing over nearly three millennia. The abundance of data fosters an environment conducive to detailed and accurate classification tasks.
- The hieroglyph recognition proposal can be applied directly to real stelae instead of isolated hieroglyphs. In this way, it makes a significant contribution to the reading and understanding of Egyptian hieroglyphic texts. Moreover, we provide accurate and relevant classification and recognition according to the context of the hieroglyphic text, since hieroglyphs do not have a definite writing direction, and one symbol can have several definitions. This opens a very wide field with which to start working on the recognition, transliteration and translation of images captured by any current device capable of capturing images with hieroglyphs.

The rest of the paper is structured as follows: Section 2 explores the state of the art related to hieroglyphic recognition and the technologies used. Section 3 describes the dataset and the proposed approach. Section 4 presents the experiments and the results obtained. Finally, Section 5 shows the conclusions and lines of future research.

2. State of the art

The task of recognizing hieroglyphs is both fascinating and difficult, partly due to the scarcity of resources. The interest in automating the reading of ancient texts has been growing in recent years thanks to the development of machine learning algorithms. Some recent applications, such as Fabricius [16], offer services for ancient language recognition. Fabricius is a collaborative project between Google Arts & Culture, the Australian Centre of Egyptology at Macquarie University, and Ubisoft. It aims to decode Egyptian hieroglyphs using Cloud AutoML [17] and three APIs: Extraction, Classification, and Translation. One of the limitations of Fabricius is the need for the user to manually delimit the hieroglyphs. Our work seeks to automate that part as well, eliminating the need for manual tagging.

Regarding the research related to the recognition of ancient characters, various techniques have been applied in order to achieve this task, such as Deep Learning with Convolutional Neural Networks (CNN) and Vision Transformers (ViT) for detection, segmentation and classification, Natural Language Processing (NLP) for analyzing texts or classic computer vision techniques, such as Histogram of Oriented Gradients (HOG), or feature extraction methods. The authors of [1] designed a system for automatic recognition of Egyptian hieroglyphs using a retrieval-based approach. The system employs a bag-of-words model to represent hieroglyphs as visual words, which are then used to build a dictionary for retrieval. The authors report promising results with a retrieval accuracy of over 95 % for a small set of hieroglyphs. The system is also scalable, as it can handle large hieroglyph datasets by using efficient retrieval techniques such as inverted indexing. However, the authors acknowledge that the system's accuracy could be improved with better feature extraction techniques and more advanced matching methods. In [18], the authors proposed an image-based Hieroglyphic Character Recognition (HCR) system based on a combination of image processing techniques. The system was evaluated on a dataset of 700 hieroglyphic characters and achieved an accuracy of 80 %. The method consisted of three stages: preprocessing, feature extraction, and classification. The preprocessing stage involves noise removal, binarization, and segmentation. The feature extraction stage uses HOG to extract feature vectors from the segmented hieroglyphic characters. Finally, a deep convolutional neural network is trained to classify the hieroglyphic characters. The proposed system shows promising results and provides a potential solution for automating the recognition of hieroglyphic characters. The approach followed by [19] is based on a combination of CNNs and a Long Short-Term Memory network (LSTM) to deal with the sequential nature of hieroglyphic texts. The authors evaluated the approach on two datasets of hieroglyphic texts, achieving high classification accuracy rates. They also demonstrated the usefulness of the approach in a case study, where they applied it to classification of hieroglyphs on a damaged papyrus fragment. The proposed method represents a promising tool for the automatic processing of ancient Egyptian hieroglyphs, with potential applications in archaeology, history, and linguistics. However, the new segmentation techniques used in our paper allow us to improve the extraction of hieroglyphs, as we will see later.

In [20], the authors proposed a method for identifying shabtis, a type

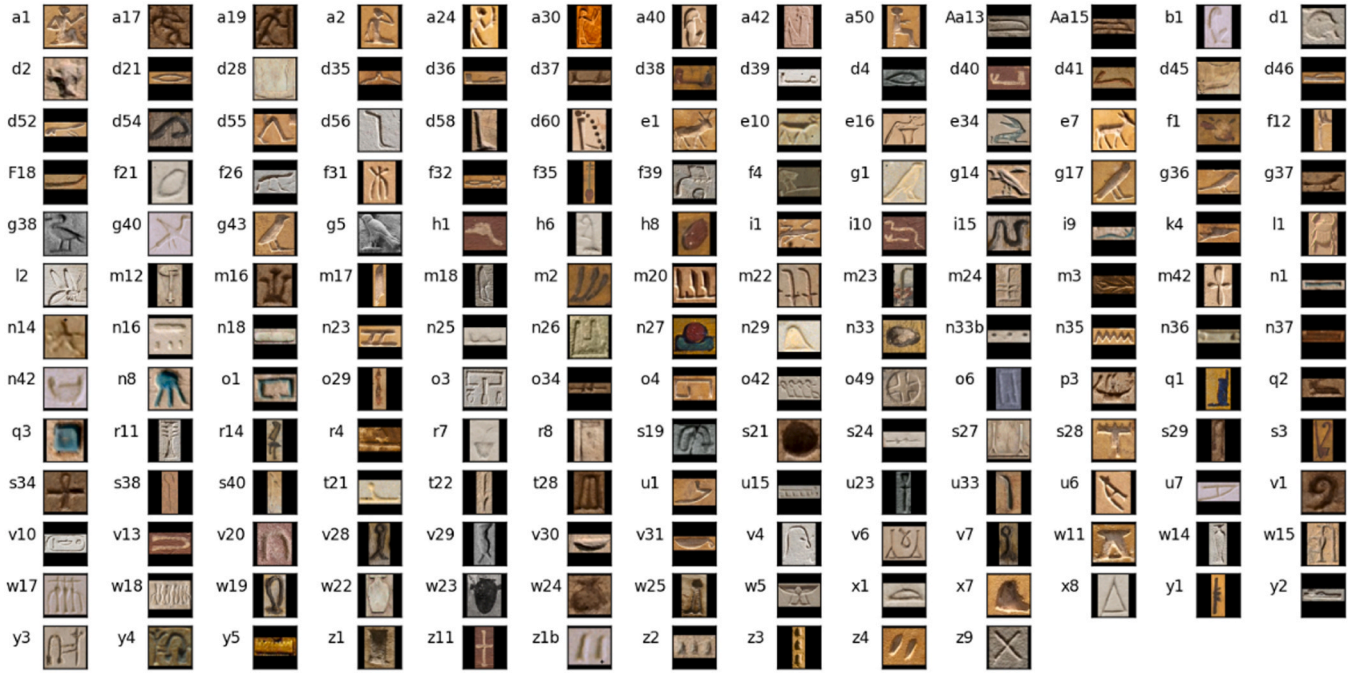


Fig. 1. Certain hieroglyph classes according to the Gardiner list.

of funerary statuette used in ancient Egypt, using deep neural networks and a semantic integration with Europeana. The shabtis were detected and classified using YOLO v3 [21] considering two aspects: the shape of the shabti and the hieroglyphic inscription. The shabtis usually have the name of the deceased person inscribed on them, and therefore YOLO was trained to search for specific names that were repeated in shabtis of the same person. The method achieved an accuracy of 90.7 % in identifying the shabtis. The authors demonstrated how semantic integration with Europeana can provide additional information about shabtis, such as their history and context, and provide an enriched interpretation.

In [22], the authors presented a text retrieval system designed to facilitate the study of ancient Egyptian texts. The system used Natural Language Processing (NLP) techniques to extract relevant information from digitized documents, including metadata, hieroglyphs, and text segments. The authors evaluated the system using a corpus of ancient Egyptian texts and found that it outperformed traditional keyword-based search methods. A probabilistic model to classify hieroglyphs in ancient Egyptian writing is presented in [3]. The model was based on Hidden Markov Models (HMMs) and used a set of features to calculate the probabilities of different hieroglyphs. The authors evaluated their model using a corpus of ancient Egyptian texts and compared its performance with other methods. The results showed that their approach outperforms other models, achieving a high accuracy in hieroglyph classification. The authors also discussed potential applications of their model in natural language processing and text recognition tasks.

In [23], the authors presented a new tool, called Scriba, for translating hieroglyphs to various languages using deep learning techniques. The authors collected a large dataset of hieroglyphic texts and trained a CNN to recognize and translate hieroglyphs into corresponding modern language words. They also developed a user-friendly web application for Scriba, allowing users to upload an image of hieroglyphic text and obtain instant translations. The results of the experiments showed that Scriba achieves high accuracy in recognizing hieroglyphs and when translating them into modern languages, with a 95 % accuracy when classifying 106 classes from the Franken dataset. In [6], an algorithm based on HOG and SURF descriptors was proposed to identify 7 classes in 41 hieroglyphic images using Euclidean distance. Their approach achieved an accuracy of 66.64 %. In [24], the authors presented an architecture for the whole hieroglyphic recognition process. However, this architecture was theoretical, and no classification or segmentation experiments were performed.

Although different from Egyptian character recognition, Mayan characters have certain similarities that merit attention. The authors of [4,25] presented a methodological approach for shape-based detection of Mayan hieroglyphs using weighted representations of a bag of words. The application of advanced image processing techniques in order to simplify the complex shapes of hieroglyphs was highlighted, thus enabling effective detection through machine learning models. This method seeks to improve the efficiency and accuracy of hieroglyphic search and retrieval in information systems by facilitating a more effective visual representation. They proposed the application of advanced image processing techniques to transform the intricate shapes of hieroglyphs into more manageable features, enabling the implementation of information retrieval models. Likewise, this approach has relevance in the interpretation and analysis of Maya hieroglyphs in archaeological and linguistic contexts, contributing significantly to the field of pattern recognition.

Another approach for Chinese character recognition is presented in [26]. In this work, an innovative deformable convolutional network model with an attention mechanism is proposed, which is specifically designed for the identification of Chinese characters belonging to various historical dynasties. This approach has the potential to be a valuable tool for archaeologists and cultural relics enthusiasts, allowing them to make rough estimates of the age of different cultural objects.

The authors of [27] propose an unsupervised learning approach for symbol recognition in document images. A feature extractor is trained on unlabeled documents and achieves remarkable accuracy rates with few reference samples. This method outperforms supervised learning approaches and is efficient in situations with little labeled data. In [28], an automatic method for detecting ancient symbols written in stone, such as stonemasons' marks, is proposed. It takes into account well-known techniques used in computer vision to identify the contour of the symbols in the image obtained under varying lighting conditions. Finally, the authors in [29] propose a method for recognizing handwritten text in resource-poor situations, such as manuscripts with rare

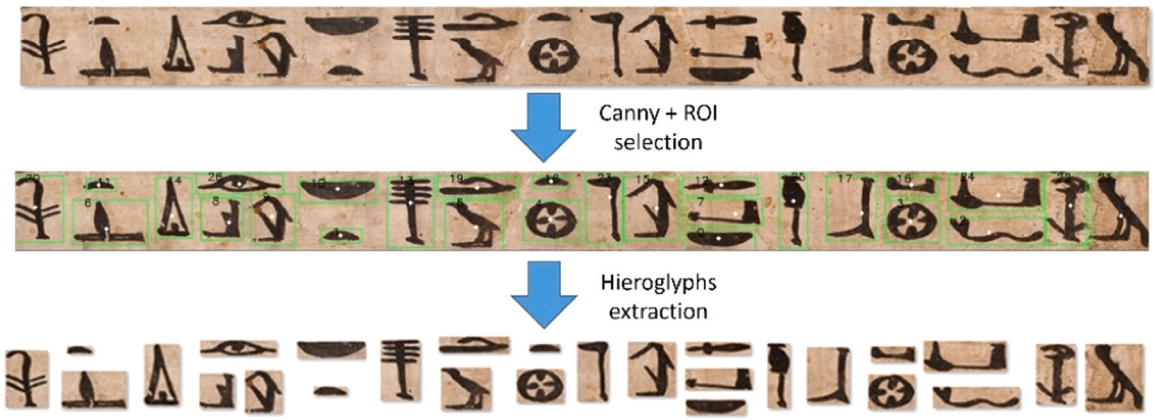


Fig. 2. Diagram of the proposed segmentation method using Canny's algorithm to extract the edges of the regions (MBRS).

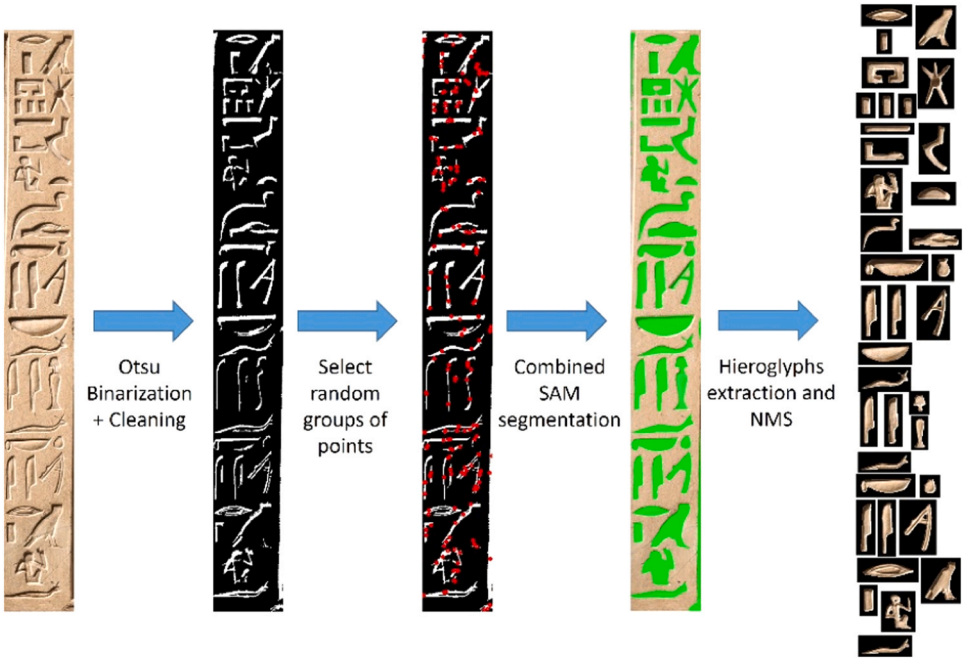


Fig. 3. Diagram of the proposed segmentation method using focused SAM with random points obtained by binarization (IGSM).

alphabets. The method requires very few examples of each alphabet, reducing the need for human annotation. It works by detecting symbols in a text image and deciphering the sequence of symbols from similarities.

These studies employ various methods to identify and interpret hieroglyphs, advancing recognition techniques and systems. Our paper presents a method that contributes to this development.

3. Proposed approach

The main objective of our system is to obtain the transliteration of a set of Egyptian hieroglyphs. Transliteration is the conversion of Egyptian hieroglyphs into their sound according to current phonetics, basically consisting of translating a set of hieroglyphs into a string of Latin alphabetic text. In our approach, we have divided the problem into two parts: segmentation and classification.

Our proposal includes the largest public dataset of Egyptian hieroglyphs, with 310 different hieroglyph classes and 13,729 images. In addition, we present two techniques for segmenting the hieroglyphs on a stele along with a combined classifier that offers high classification accuracy. We will first present the dataset and then the segmentation and classification techniques.

3.1. Dataset

We have created a dataset with 310 classes and 13,729 images, representing hieroglyphs on different materials. Our dataset consists of images labeled by us together with images from the Franken dataset [1]. Specifically, we have manually classified 9703 images into 310 classes. To the best of our knowledge, it is the largest public dataset currently available, larger than that proposed by Franken with 168 classes and 4026 images and by Barucci et al. [19] with 48 classes and 1310 images. The hieroglyphs have been grouped in folders according to the structure and typology defined by the Egyptologist Gardiner [30,31] (see Fig. 1). The name of the class is made up of an alphabetic character followed by a number. The Gardiner code is a unique identifier for each hieroglyph class.

The images obtained have been classified and labeled for each of Gardiner's classes. The images were obtained from various sources: museum stelae, handwritten or sculpted, carved, etc., and belong to

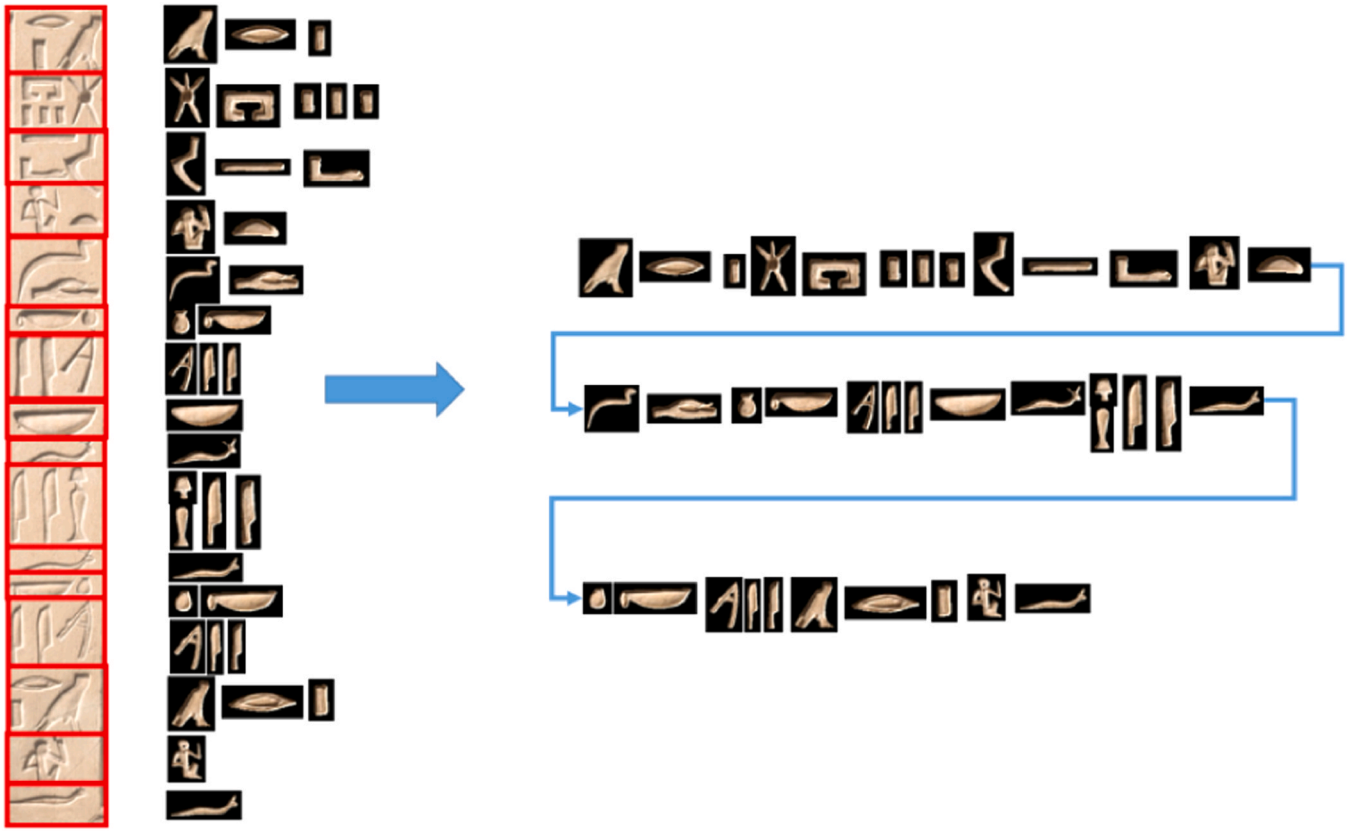


Fig. 4. Organizing the hieroglyphs so they can then be read.

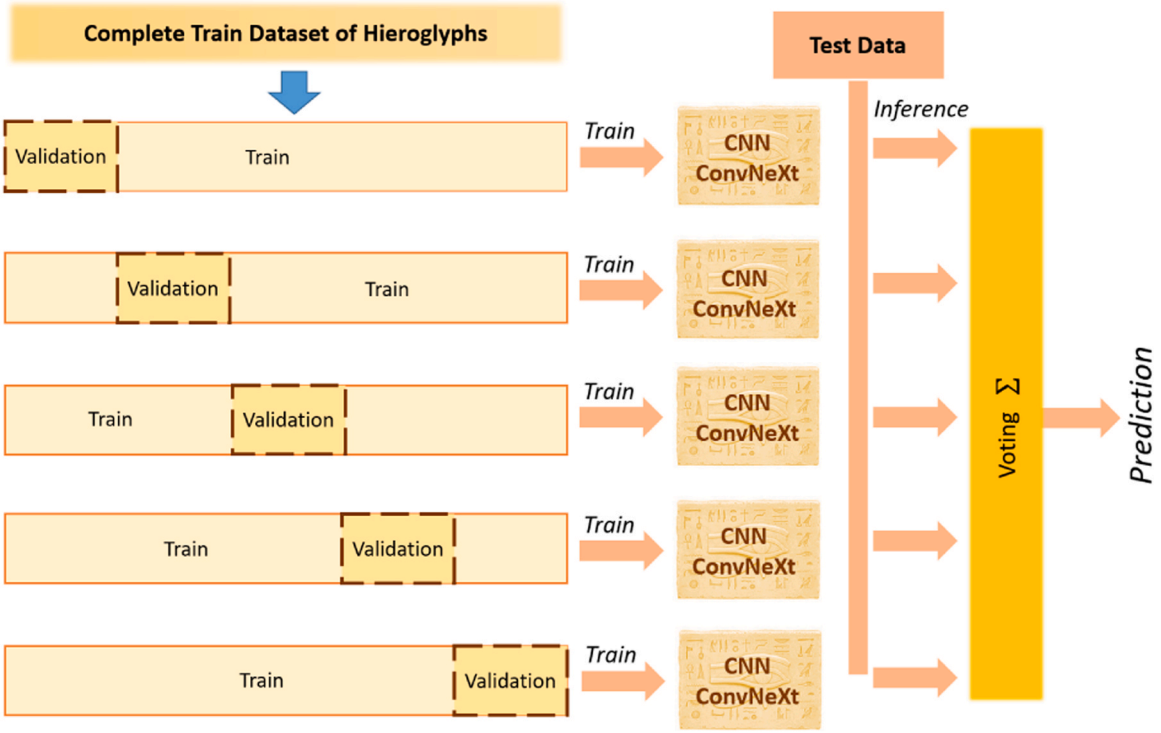


Fig. 5. Flowchart of the proposed classification method using CVV.

different periods of Egyptian history, providing a wide generalization at the time of recognition. In addition, the images obtained have different backgrounds and sizes, so a preprocessing has been performed so that all the images have the same dimensions in X or Y. The images have been cropped from scanned photos, with a rectangular shaped border, thus obtaining a uniform format for each image.

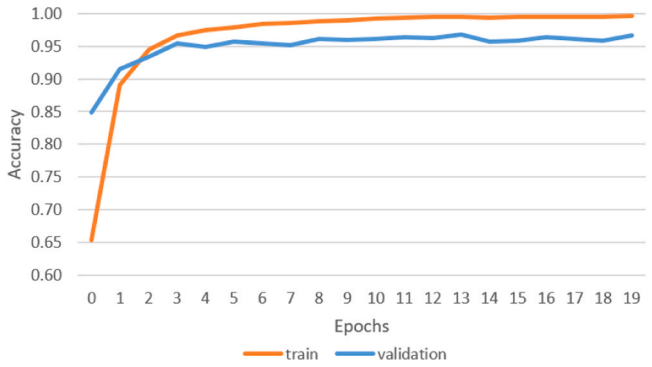


Fig. 6. Accuracy of CVV slot 0.

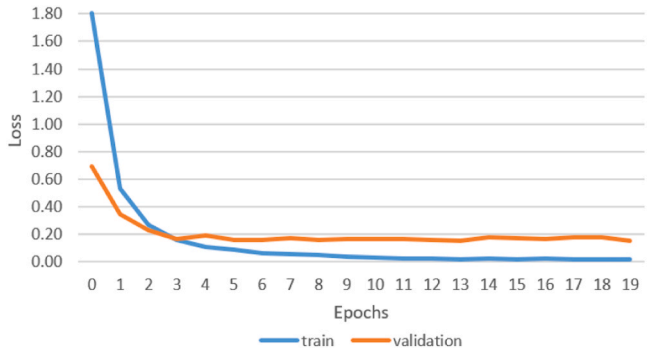


Fig. 7. Loss of the CVV slot 0.

Table 1
Results of the various tests performed on the models trained with the different slots (MSLOT0 to MSLOT4), and those obtained with the CVV in its two variants studied: Soft Voting (CVV-SV) and Hard Voting (CVV-HV). The best results are highlighted in bold.

	Model	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Sensitivity (Recall or TPR)*	*F1 score*
TEST	MSLOT0	0.928	0.839	0.804	0.795	0.799
	MSLOT1	0.933	0.841	0.801	0.797	0.799
	MSLOT2	0.928	0.856	0.812	0.811	0.811
	MSLOT3	0.926	0.842	0.803	0.798	0.800
	MSLOT4	0.926	0.837	0.795	0.793	0.794
	CVV-SV	**0.960**	**0.917**	**0.917**	**0.917**	**0.917**
	CVV-HV	0.958	0.910	0.907	0.910	0.908

3.2. Segmentation

Segmentation is a process that can be implemented using classical vision techniques or deep learning models. Specific segmentation models, such as Mask R-CNN [32] or Detectron2 [33], require manual labeling of hieroglyphic boundaries, which is too laborious. We propose two variants of segmentation, one based on classical vision techniques and the other using focused SAM [11], designed to segment any type of object without the need for training. It is a kind of zero-shot learning segmentation model.

In addition to the segmentation techniques, and before implementing the proposed solution, we tested various detection methods, such as YOLO networks [34]. However, the labeling in these methods is very expensive and requires many bounding boxes around hieroglyphs with different backgrounds in reliefs or stelae, composed of other hieroglyphs. In our case, we did some quick tests with YOLO [34] before continuing with our method, but the results were not accurate. For this reason, we discarded it.

We have evaluated two segmentation methods. The first method is based on classic computer vision techniques. It uses Canny's algorithm [35] to extract the edges of the regions. We have designated the approach we have developed as a Method Based on Region Segmentation (MBRS). Once the edges are detected, we analyze the distinct regions using the edges that are separated. Using this technique, we can correctly isolate cases such as the example shown in Fig. 2, where there is a considerable difference between the color of the hieroglyphs and that of the image background itself. We then select the bounding box that circumscribes the independent border, and separate each hieroglyph into a separate image (green rectangles in Fig. 2). The separated images are later processed by CVV as described in Section 3.3, a technique that uses a combined classifier trained with different validation sets. The CVV technique then returns the classes of the hieroglyphs, which correspond to a Gardiner code and a specific transliteration. Finally, from the transliteration we can reconstruct the hieroglyphs using JSesh [15], and we could even carry out the translation using some kind of transformation model.

The second method (see Fig. 3), an alternative intended for cases where edges are harder to define, uses focused general segmentation based on SAM [11]. We have named this methodological approach Integrated Generic Segmentation Model (IGSM). Applying SAM directly to the images without any prompts returns segmentation masks mixed between hieroglyphs. In addition, we found that there were many overlapping detections that were difficult to deal with. For this reason, we used point segmentation, proposing an approach to obtain the key points. Our method starts with a binarization of the image using Otsu's method. This method returns areas of the image with shadows, such as gaps or hieroglyphic edges. We then remove very small areas that may be due to noise. For each connected region of points, we select between 8 and 11 points at random and perform a per-point segmentation using SAM. According to our experiments, we found that SAM does not always perform the hieroglyph segmentation well, leaving uncovered areas. By selecting several random points and combining the segmented detections, the hieroglyph improved significantly. More specifically, the number of random points used by SAM is dependent on the material typology employed: between 6 and 11 points are used for hieroglyphs carved in stone, while between 3 and 8 points are used for hieroglyphs painted on stone. We also use a non-maximum suppression (NMS) [36] algorithm to avoid finding the same hieroglyph twice if there is some separation in the binary zones selected with Otsu. We also reject hieroglyphs with a size above certain defined limits. They could be due to broken parts of the stele. Once the hieroglyphs have been located, we are able to segment them. Our method can exclusively extract the hieroglyph, rejecting the rest of the background (see Fig. 3). This is particularly useful for rejecting hieroglyphs that are close to other hieroglyphs, which could cause classification problems. However, in our experiments, we found that the classification performed better on images with the segmented hieroglyph and the rest of the background. Therefore, we select a bounding box around the segmented hieroglyph including the background for the classification task.

Along with the segmentation, the hieroglyphs must be organized so they can then be read. We assume a reading from right to left and from top to bottom. If the image corresponds to a column (taller than wide), when a large hieroglyph appears on the right and one or more smaller ones on the left, we consider that the reading starts with the hieroglyph on the right, followed by the smaller ones from top to bottom. The next blocks are then read, as shown in Fig. 4. The Egyptians placed their hieroglyphs individually or grouped in blocks, as shown in the red rectangles in the figure.

3.3. Classification

In our proposal, we have given considerable importance to the generalizability of the classification, and have therefore opted for the

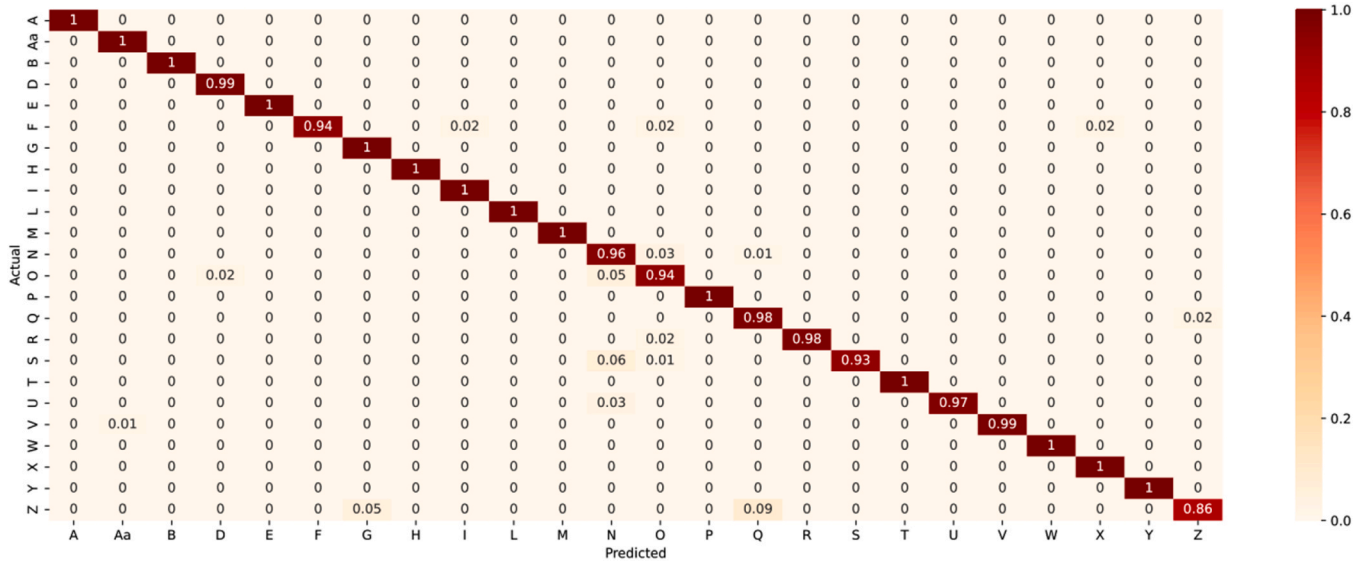


Fig. 8. Confusion matrix of soft voting CVV with ConvNeXt.

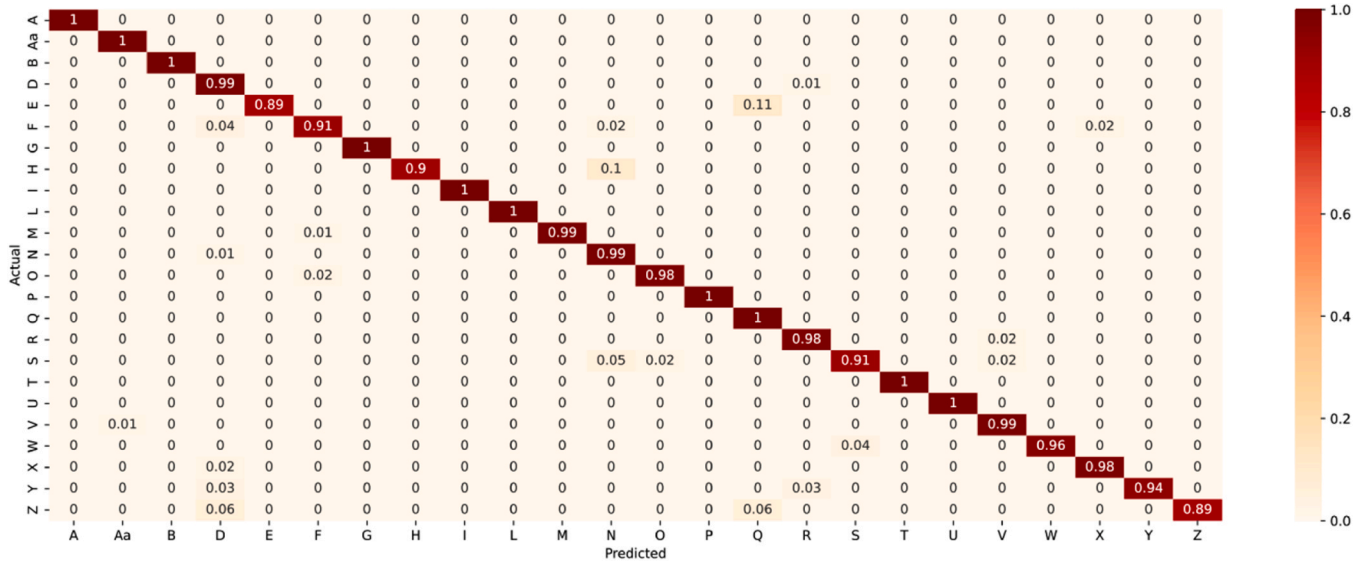


Fig. 9. Confusion matrix of hard voting CVV with ConvNeXt.

CVV classification technique [37], which allows the use of different similar classifiers trained on disjoint validation groups. We have used the ConvNeXt backbone [13]. Unlike traditional methods that select random data sets to train different models, in our method we use validation sets to avoid overfitting. This is especially relevant in neural networks, where early stopping mechanisms allow us to cease training before overfitting occurs. The CVV method involves distributing the training data set into different training and validation slots. In this way, all the data are used to train the joint model, where each individual model uses one part of the training data set as training and another as validation. This ensures that no samples are lost during the training process. Fig. 5 shows the CVV technique applied to the classification of Egyptian hieroglyphs.

In the CVV method, the data are first randomized and then k distinct validation slots are selected. The complete training data set is denoted as τ. For each slot i, the corresponding training and validation sets are denoted as T_i and V_i, respectively. These sets must comply with Eqs. 1 to 4.

$$\tau = \bigcup_{i=1}^{k} V_i \tag{1}$$

$$\bigcap_{i=1}^{k} V_i = \varnothing \tag{2}$$

$$[\tau = T_i \cup V_i] \forall i \in [1, k] \tag{3}$$

$$[T_i \cap V_i = \varnothing] \forall i \in [1, k] \tag{4}$$

Once the models have been trained with each set of validation and training slots, they are combined using classical voting techniques. We denote by k the number of classifiers, each associated with its corresponding validation slot. Given an input sample x, $\overrightarrow{p_{i(x)}}$ is the vector of output probabilities delivered by classifier i. This vector is composed of the probabilities $p_{ij}(x)$, which indicate that sample x belongs to class j according to classifier i. The label vector $\overrightarrow{c}$ consists of a label for each possible class: $\{c_1, c_2, \ldots, c_n\}$. In Eq. (5), the soft vote $S(x)$ is obtained by

Table 2
Class grouping according to Gardiner's typology.

Coarse Class	Description	Fine Classes
A	Man and his Occupations	A1, A2, A19, A24, A30, A40, A42, A50
B	Woman and her Occupations	B1
D	Parts of the Human Body	D1, D2, D4, D21, D28, D35, D36, D37, D39, D40, D45, D46, D52, D54, D55, D56, D58, D60
E	Mammals	E1, E7, E21, E23, E34
F	Parts of Mammals	F1, F4, F12, F13, F18, F21, F26, F31, F32, F34, F35, F39,
G	Birds	G1, G5, G7, G14, G17, G25, G35, G36, G37, G38, G39, G40, G43
H	Parts of Birds	H1, H6, H8
I	Amphibious Animals, Reptiles, etc.	I1, I9, I10
L	Invertebrate and Lesser Animals	L1, L2
M	Trees and Plants	M2, M3, M12, M16, M17, M18, M20, M22, M23, M42
N	Sky, Earth, Water	N1, N5, N8, N14, N18, N25, N26, N29, N30, N31, N33, N35, N36, N37, N42
O	Buildings, Parts of Buildings, etc.	O1, O3, O4, O6, O28, O29, O34, O49, O50
P	Ships and Parts of Ships	P8
Q	Domestic and Funerary Furniture	Q1, Q2, Q3
R	Temple Furniture and Sacred Emblems	R4, R7, R8, R11, R14
S	Crowns, Dress, Staves, etc.	S3, S19, S21, S24, S27, S28, S29, S34, S38, S40, S42
T	Warfare, Hunting, Butchery	T21, T22, T28
U	Agriculture, Crafts and Professions	U1, U6, U7, U15, U23, U33
V	Rope, Fiber, Baskets, Bags, etc.	V1, V4, V6, V7, V10, V13, V20, V28, V29, V30, V31
W	Vessels of Stone and Earthenware	W11, W14, W15, W17, W18, W19, W22, W23, W24, W25
X	Loaves and Cakes	X1, X7, X8
Y	Writings, Games, Music	Y1, Y2, Y3, Y4, Y5,
Z	Strokes, Signs derived from Hieratic, Geometrical Figures	Z1, Z2, Z3, Z4, Z11
Aa	Unclassified	Aa13, Aa15

accumulating the output probabilities of each class j. A weight w_i is assigned to each classifier i; in our case all weights are $\frac{1}{k}$. The argmax function returns the class position with the highest cumulative probability.

$$S(x) = c_{\arg_j \max} \sum_{i=1}^{k} w_i \cdot p_{ij}(x) \tag{5}$$

For hard voting, the probability needs to be binarized beforehand, which is represented in Eq. (6). To achieve this, only the class with the highest probability is set to 1, while the rest are set to 0.

$$b_{ij}(x) = \begin{cases} 1 \text{ if } p_{ij}(x) = \max(\overrightarrow{p_i(x)}) \\ 0 \text{ otherwise} \end{cases} \tag{6}$$

Eq. (7) obtains the output $H(x)$ by summing the binary values corresponding to each j class. As in soft voting, w_i is taken to be equal to $\frac{1}{k}$ in our case.

$$H(x) = c_{\arg_j \max} \sum_{i=1}^{k} w_i \cdot b_{ij}(x) \tag{7}$$

Random data augmentation was used in the training to achieve a better balance of the training set. New elements were added to classes containing a number of images below a threshold, while images were removed from classes containing too many images. Specifically, random translation along both horizontal and vertical directions by a maximum of 10 pixels, small random rotations of 5 degrees, and random scaling with a factor of 0.90–1.10 were used. Next, since hieroglyphs can be oriented either to the left or to the right (without changing their meaning), horizontal flipping was also used. Finally, the maximum number of training images that a class can contain was set to twice the average number of images per label. The total number of images in the training set was 9359 and, after data augmentation, 18,718.

4. Experiments and results

In our experiments, we utilized 13,169 hieroglyphs, categorized into 164 classes, by setting a minimum threshold for the number of hieroglyphs per class of 7 to improve result reliability. The classification dataset is divided into 90 % for training and validation, and 10 % for testing. Within the training and validation set, the CVV method further divides the 90 % into 70 % training and 20 % validation groups. Specifically, during a CVV training we will have 11,773 images on which the data augmentation, described in Section 3.3 is performed, until 18,718 images are obtained, divided into 14,558 training images and 4160 validation images. The test set remains at 1396 images. In total we used 13,169 images during CVV training and evaluation.

This section presents the results of the proposed segmentation and classification methods performed on the dataset. First, the classification results with and without CVV are presented. Second, the results of the segmentations integrated with the classifiers are presented. Finally, a comparison with previous recent works is included, based on the characteristics of the dataset used and the proposed techniques.

4.1. Individual classification results

Fig. 6 shows the accuracy values corresponding to the training process with the 0 slot used in this work. We see that the accuracy is close to 1, its closest value being 0.967. Fig. 7 shows the loss of the process in the 0 slot. There are slight variations in the loss depending on the slot used, with slot 0 being the one that also gives the best results with a value close to 0.150. The best validation values in both cases (accuracy and loss) correspond to the epoch number 13.

Table 1 shows the values corresponding to the various tests performed on the test folder contents, both for the models trained with the different slots (0−4) and those obtained with the CVV in its two variants studied (Soft and Hard Voting). Confusion matrices corresponding to the CVV classification results have been generated using both Soft (Fig. 8) and Hard Voting (Fig. 9) methods in order to meticulously examine the precision of the predictions based on the selected approach. However, due to the large number of thin classes (164), we present the results of the classes grouped according to Gardiner's typologies. Table 2 shows how these hieroglyphs are grouped according to the typologies, resulting in a total of 24 categories. This categorization facilitates visual comprehension. In this context, Soft Voting provides a more precise approximation than Hard Voting.

4.2. Individual segmentation results

With respect to the segmentation proposal, a detailed analysis of various stelae was performed, covering a wide range of materials and media and taking both into account in the dataset to ensure a comprehensive understanding. Here we show one of the evaluated stelae corresponding to the Maati Gate guardian (Fig. 10).

In the context of this research, the extraction of the top row of the stela was undertaken, as illustrated in Figs. 11 and 12, together with a careful analysis of four columns, as depicted in Figs. 13 and 14. This extraction process yielded a set of five autonomous images, each of which was subjected to an inversion procedure. This inversion step was undertaken with the specific purpose of improving the positional recognition of individual hieroglyphs at the time of their subsequent extraction, which is considered essential in the process of analysis and classification of hieroglyphic elements on Egyptian stelae.



Fig. 10. Stela of the Gatekeeper Maati (Courtesy of the MET).



Fig. 11. Row extracted from the Stela of the Gatekeeper Maati.

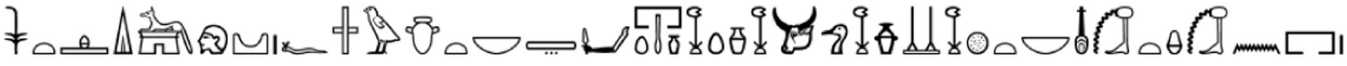


Fig. 12. Corresponding hieroglyphic representation with JSesh.

The extraction process was designed with the aim of obtaining individualized hieroglyphs present in the images analyzed. As mentioned above, two specific techniques were used to carry out this task: the Method Based on Region Segmentation (MBRS) and the Focused Integrated Generic Segmentation Model (IGSM). The MBRS extraction process is shown in detail in Fig. 15, and the IGSM extraction process is shown in Fig. 16. Both figures illustrate the individual extraction of each of the hieroglyphs and subsequent classification based on morphological and structural features.

It is important to note that the fit and accuracy of the extraction process varies from image to image, depending on the complexity of the hieroglyphic layout. In order to evaluate the quality of the extraction, a classification process was carried out using the Cross-Validation Voting (CVV) technique, both in its soft and hard voting modes. These techniques allowed researchers to determine the adequacy of the segmentation and the accuracy of the classification. Finally, in an integral step of the process, the JSesh representation of the extracted hieroglyphs is presented, which completes the analysis and validation phase of the extraction, as described in the present study. These procedures, supported by rigorous evaluation and classification, generated robust results in the automated interpretation and reading of Egyptian hieroglyphic texts.

In both situations, it is observed that there are hieroglyphs that are not perfectly recognized when compared to their actual representation. This leads us to consider, in the classification process, the possibility that some of the hieroglyphs we are trying to classify are not found in the dataset. We can then perform the analysis in two different ways: the first involves classification of all the hieroglyphs obtained in the extraction process, regardless of whether they are contained in the current dataset; the second focuses only on classification of those hieroglyphs included in the dataset. The choice between these two analytical methods will depend on the specific research objectives and the practical needs of the system. This flexibility in the classification approach is critical in ensuring that the results obtained are consistent with data availability and the long-term goals of developing automated methods for interpreting Egyptian hieroglyphic texts.

4.3. Results of the combined segmentation and classification proposal (MBRS-CVV and IGSM-CVV)

As discussed above, the main results of the evaluation of the two hieroglyph segmentation methods proposed, the MBRS and the IGSM, are presented focusing on the classification of all the hieroglyphs and on the classification of only the hieroglyphs present in the dataset. In the latter case, the IGSM showed even better performance, with accuracies ranging from 0.90 to 0.97. Balanced accuracy remained high, with values ranging from 0.94 to 0.99. Recall was consistent, exceeding 0.90 in most cases, and the F1 score remained around 0.91. The overall



Fig. 13. Columns extracted from the Stela of the Gatekeeper Maati.

evaluation showed an accuracy of 0.95 and an F1 score of 0.91. The MBRS showed a more modest performance compared to the IGSM, with precision, balanced accuracy, recall and F1 score values of a range between 0.50 and 0.86.

In the analysis, we also included the totality of the hieroglyphs present in the analyzed stele in order to obtain a global understanding of the study. For the CVV-SV approach, the IGSM showed a solid performance with accuracies ranging from 0.81 to 0.94. The balanced accuracy remained high, with values ranging from 0.89 to 0.95. Recall was consistent, exceeding 0.77 in most cases, and the F1 score remained around 0.81. The overall evaluation (Complete Stela) showed an accuracy of 0.89 and an F1 score of 0.80. In the CVV-HV approach, the results were largely similar in terms of balanced accuracy, precision, recall and F1 score, although some variations in specific values were observed. These observations support the effectiveness of the proposed approach in the classification of Egyptian hieroglyphs, which is of key importance in the field of automated interpretation of hieroglyphic texts. Tables 3–6 show the classification results of the MBRS and IGSM for CVV-SV and CVV-HV for the row and columns of the Stela of Maati (Fig. 10).

In summary, the IGSM shows promising performance in hieroglyph classification, particularly when considering only the hieroglyphs present in the dataset, and the results support its usefulness in the automated reading of Egyptian hieroglyphs. In comparison, MBRS shows a more modest performance. These findings are highly material regarding the advancement of the field of Egyptology and hieroglyphic analysis.

In addition, a comprehensive verification of a complete set of four hieroglyphic inscriptions from different historical periods was performed to cover the nearly 3000 years of development of known hieroglyphic writing. The results of this analysis are presented in Table 7. This analysis evaluates the totality of the hieroglyphs found on each of the stelae analyzed. Overall, the accuracy achieved by the IGSM method on the stelae gives superior results compared to the MBRS. The results of this analysis provide a solid foundation for future research in the field of Egyptology and automated interpretation of hieroglyphic texts.

4.4. Comparison with state-of-the-art datasets and hieroglyph recognition approaches

Finally, in Table 8 we compare our dataset with those proposed by [1,19,10], which we refer to as the Franken, Barucci, and Sobhy datasets respectively. Each of these datasets has unique characteristics that are crucial for their applicability in this type of research. One of the most important aspects to consider is the historical breadth of each dataset. The first is limited to a single period, the Old Kingdom, and provides a detailed but limited view of hieroglyphs in a specific time period. In contrast, our dataset excels in encompassing all periods of ancient Egyptian history, the Old, Middle, and New Kingdoms, which is essential for a comprehensive understanding of the evolution of hieroglyphic writing over nearly three millennia. This approach provides a holistic perspective that few datasets can match.

In their respective research, [1,19] propose hieroglyph recognition approaches based on detection models. Franken and van Gemert [1] use the HOG descriptor with a single descriptor matching scheme. Barucci et al. [19] use the Detectron2 network for the segmentation [33] and a proprietary model called Glyphnet for the classification. Both studies evaluate their methods using the Average Precision (AP) score for the detection of hieroglyphs in stelae, specifically considering an Intersection over Union (IoU) threshold of 50 % (AP50). In the case of Sobhy et al. [10], an R-CNN [38] detection model is used together with a Siamese network-based classifier. The authors of [9] use a Mask R-CNN segmentation network and present AP results in the validation set, reaching a value of 0.93. However, it is important to note that this work



Fig. 14. Hieroglyphic representation with JSesh of the columns extracted from the stela (reading from left to right); the top row corresponds to the right column, and so on.

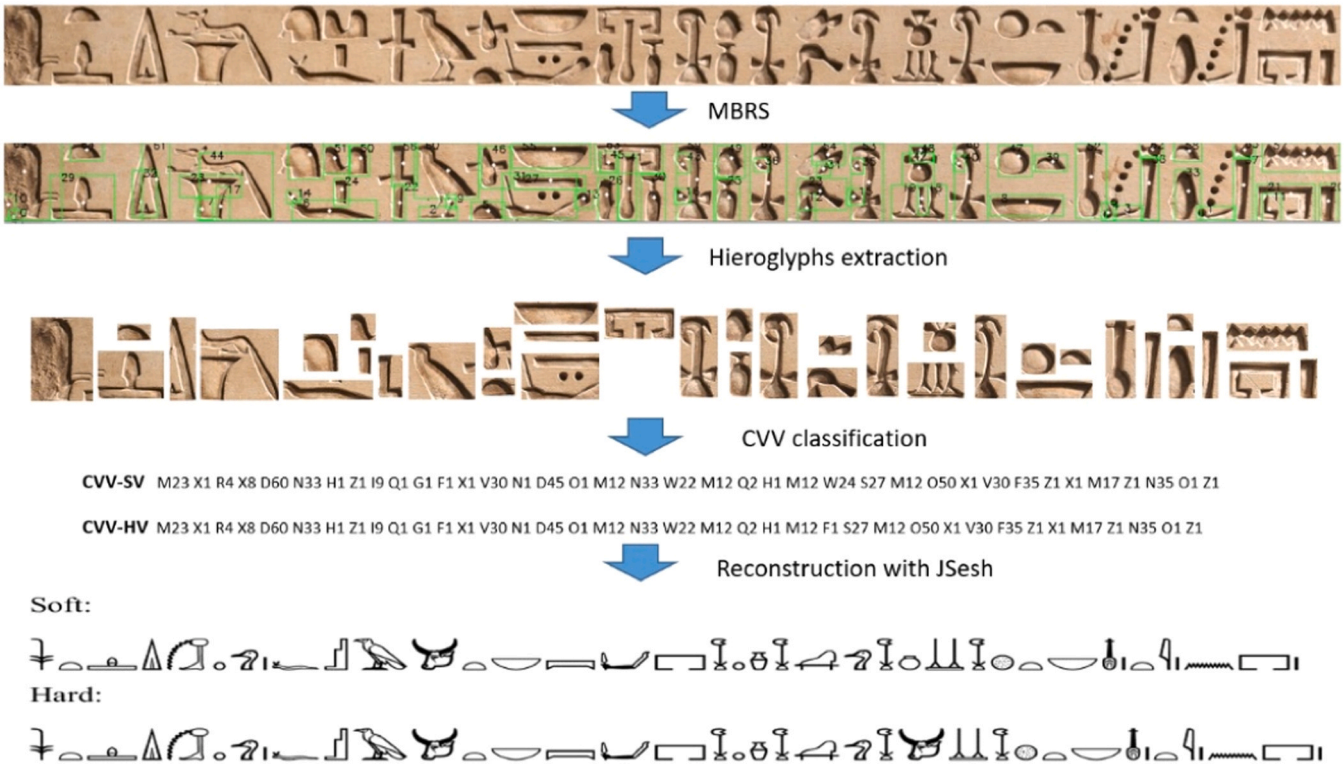


Fig. 15. Flowchart of the proposed process based on MBRS and CVV.

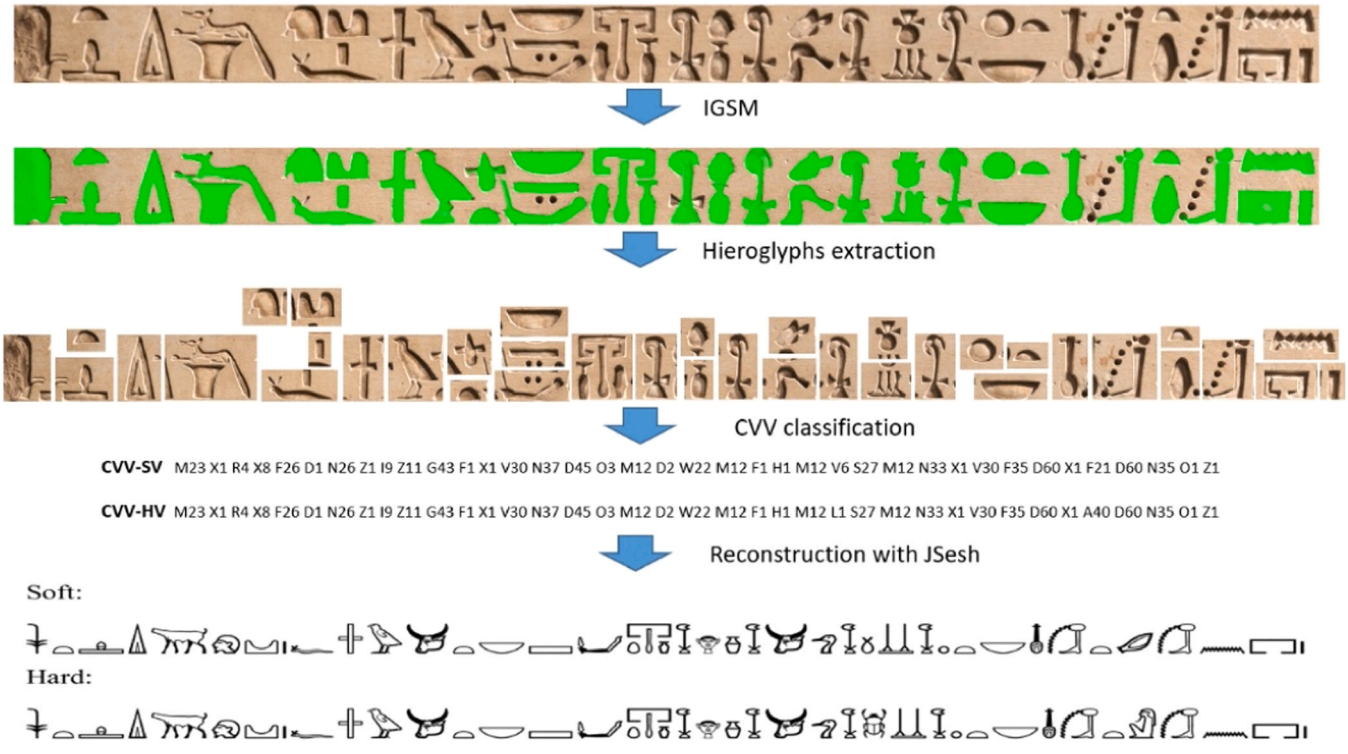


Fig. 16. Flowchart of the proposed process based on IGSM and CVV.

does not provide quantitative results regarding the test set. Table 8 also compares our proposed segmentation-classification approach with those proposed in [1,19,10].

The richness of a dataset in terms of hieroglyphs and classes directly influences its versatility. Our dataset stands out with 13,729 hieroglyphs and 310 classes. This abundance fosters an enriched research environment, which is essential for detailed and accurate classification tasks. In contrast, the Franken and Barucci datasets possess more modest amounts in both respects, which may limit their applicability in research seeking broad hieroglyphic diversity.

Table 3
MBRS: Method based on Region Segmentation - all hieroglyphs in stela.

CVV-SV	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Recall (sensitivity or TPR)*	*F1 score*
ROW 1	0.605	0.625	0.422	0.469	0.444
COLUMN 1	0.794	0.740	0.638	0.638	0.638
COLUMN 2	0.833	0.753	0.622	0.622	0.622
COLUMN 3	0.606	0.600	0.451	0.444	0.447
COLUMN 4	0.722	0.686	0.550	0.581	0.565
COMPLETE	0.712	0.636	0.523	0.556	0.539
CVV-HV	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Recall (sensitivity or TPR)*	*F1 score*
ROW 1	0.605	0.625	0.435	0.484	0.458
COLUMN 1	0.824	0.780	0.696	0.696	0.696
COLUMN 2	0.833	0.753	0.622	0.622	0.622
COLUMN 3	0.667	0.700	0.564	0.538	0.551
COLUMN 4	0.722	0.686	0.550	0.581	0.565
COMPLETE	0.729	0.626	0.539	0.581	0.559

Table 4
MBRS: Method based on Region Segmentation - only hieroglyphs in Dataset.

CVV-SV	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Recall (sensitivity or TPR)*	*F1 score*
ROW 1	0.719	0.652	0.482	0.536	0.508
COLUMN 1	0.818	0.771	0.679	0.661	0.670
COLUMN 2	0.857	0.794	0.681	0.681	0.681
COLUMN 3	0.645	0.632	0.507	0.500	0.503
COLUMN 4	0.743	0.719	0.596	0.629	0.612
COMPLETE	0.759	0.647	0.547	0.572	0.559
CVV-HV	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Recall (sensitivity or TPR)*	*F1 score*
ROW 1	0.719	0.652	0.500	0.556	0.527
COLUMN 1	0.848	0.812	0.741	0.722	0.731
COLUMN 2	0.857	0.794	0.681	0.681	0.681
COLUMN 3	0.710	0.737	0.638	0.609	0.623
COLUMN 4	0.743	0.719	0.596	0.629	0.612
COMPLETE	0.777	0.667	0.564	0.598	0.581

Table 5
IGSM: Integrated Generic Segmentation Method - all hieroglyphs in stela.

CVV-SV	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Recall (sensitivity or TPR)*	*F1 score*
ROW 1	0.816	0.917	0.717	0.733	0.725
COLUMN 1	0.941	0.950	0.904	0.913	0.908
COLUMN 2	0.944	0.895	0.810	0.810	0.810
COLUMN 3	0.848	0.892	0.792	0.743	0.767
COLUMN 4	0.917	0.909	0.833	0.833	0.833
COMPLETE	0.893	0.922	0.782	0.816	0.799
CVV-HV	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Recall (sensitivity or TPR)*	*F1 score*
ROW 1	0.816	0.917	0.717	0.733	0.725
COLUMN 1	0.941	0.950	0.904	0.913	0.908
COLUMN 2	0.944	0.895	0.810	0.810	0.810
COLUMN 3	0.848	0.892	0.804	0.775	0.789
COLUMN 4	0.917	0.909	0.833	0.833	0.833
COMPLETE	0.893	0.922	0.799	0.840	0.819

The choice of an appropriate dataset is essential for achieving research objectives. The Franken, Barucci, and Sobhy datasets may be appropriate for research that focuses on specific historical periods or hieroglyphic styles. However, our dataset is a solid choice for tasks that require a panoramic view of the development of hieroglyphic writing in Egyptian history. Its historical breadth and diversity of hieroglyphs and classes make it an excellent choice for comprehensive research and analysis throughout the Pharaonic kingdom.

Table 6
IGSM: Integrated Generic Segmentation Method - only hieroglyphs in Dataset.

CVV-SV	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Recall (sensitivity or TPR)*	*F1 score*
ROW 1	0.969	0.957	0.917	0.917	0.917
COLUMN 1	0.970	0.990	0.979	0.990	0.984
COLUMN 2	0.971	0.944	0.895	0.895	0.895
COLUMN 3	0.903	0.939	0.905	0.849	0.876
COLUMN 4	0.943	0.952	0.909	0.909	0.909
COMPLETE	0.952	0.937	0.895	0.922	0.908
CVV-HV	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Recall (sensitivity or TPR)*	*F1 score*
ROW 1	0.969	0.957	0.917	0.917	0.917
COLUMN 1	0.970	0.990	0.979	0.990	0.984
COLUMN 2	0.971	0.944	0.895	0.895	0.895
COLUMN 3	0.903	0.939	0.905	0.849	0.876
COLUMN 4	0.943	0.955	0.909	0.909	0.909
COMPLETE	0.952	0.937	0.895	0.922	0.908

5. Conclusions

This study represents a unique and unprecedented contribution to the recognition of Egyptian hieroglyphs using contemporary segmentation and classification methods. First and foremost, it involves the creation of an extensive and comprehensive dataset encompassing existing Egyptian hieroglyphs, with a particular focus on those frequently observed on stelae. This large-scale dataset provides a solid and rich foundation for future research in the field. Second, it includes a rigorous and systematic review of existing work in the field and an analysis of current detection, segmentation, and classification methods. After evaluation of detection methods and noting their unsatisfactory results, a refined approach has been chosen that focuses on generic segmentation using an own method integrated with SAM and an accurate classifier. The use of a joint ConvNeXt backbone model and the application of the Cross-Validation Voting (CVV) technique have led to excellent results in hieroglyphic classification. The evaluation of our model has demonstrated its exceptional performance on a variety of stone stelae, whether carved or painted. The results strengthen the confidence in the effectiveness and accuracy of the proposed method. Finally, this study and the methodology presented, together with the dataset developed, represent a significant contribution to the field of automated methods for reading and understanding Egyptian hieroglyphic texts. The classification results obtained are by far the most promising to date, confirm the importance of this study as a key reference in the automated translation of the ancient Egyptian script, and lay the groundwork for future advances in this area.

In future research, we envision a wide range of opportunities for improving and enriching our current approach to Egyptian hieroglyphs recognition. One of the most important aspects will be the expansion and diversification of the dataset by including a larger number of hieroglyphs. This will allow a greater representativeness of hieroglyphic writing and, consequently, an improvement in the robustness and generalization of our model. In addition, there is a need to expand the variety of classes used to train our model, which will allow us to more accurately deal with the complexity and diversity of hieroglyphs, thus optimizing the model's ability to recognize and classify different shapes and meanings. This expansion of classes will also be supported by an ongoing effort to collect and organize information on the stelae, including different typologies of writing such as painted, relief, and engraving, with the aim of improving the adaptability and effectiveness

Table 7
Averaged results obtained with the two proposed segmentation methods (MBRS and IGSM) and the classification proposal methods (CVV-SV and CVV-HV) over of all the stelae processed: (1) all the hieroglyphs in the stelae; (2) only the hieroglyphs in the dataset.

	CVV	Dataset	*Accuracy*	*Balanced Accuracy*	*Precision (or PPV)*	*Sensitivity (Recall or TPR)*	*F1 score*
MBRS	SV	ALL (1)	0.673	0.657	0.507	0.547	0.523
		ONLY (2)	0.740	0.683	0.583	0.607	0.593
	HV	ALL (1)	0.677	0.660	0.520	0.530	0.533
		ONLY (2)	0.743	0.683	0.590	0.610	0.600
	CVV	**Dataset**	***Accuracy***	***Balanced Accuracy***	***Precision (or PPV)***	***Sensitivity (Recall or TPR)***	***F1 score***
IGSM	SV	ALL (1)	0.867	0.893	0.770	0.800	0.783
		ONLY (2)	0.940	0.917	0.887	0.893	0.887
	HV	ALL (1)	0.867	0.897	0.780	0.807	0.790
		ONLY (2)	0.923	0.907	0.860	0.870	0.863

Table 8
Summary of key statistics for each dataset and comparison of test accuracy (%) with other current models.

Model	Dataset	Hieroglyphs	Classes	Accuracy	Precision	Recall	F1-Score
HOG+SDM [1]	*Franken*	*3993*	*161*	*0.855*	-	-	-
Detectron2 +Glyphnet [19]	*Barucci*	*4309*	*40*	*0.934*	-	-	-
R-CNN+SiameseNN [10]	*Sobhy*	*4210*	*172*	*0.885*	-	-	-
IGSM+CVV-SV	*Our*	*13,169*	*164*	*0.940*	*0.887*	*0.893*	*0.887*

of the model in different media. In addition, the intention is to deepen the detection aspect, addressing both segmentation and classification, on types of media other than stelae. Research will be oriented towards the accurate detection of hieroglyphs in different manifestations, such as on manuscripts, papyri, and other archaeological objects, exploring the adaptability of the model to different contexts and conditions. The translation from Gardiner codes into modern languages, such as English, would also be addressed. In summary, future lines of research will focus on the expansion and diversification of the dataset, the extension of the training classes, the inclusion of diverse typologies of writing on stelae, the deepening of hieroglyphic detection in different media, and the final translation into modern languages, thus enriching our current approach and strengthening its applicability in a wide range of scenarios related to the automated reading of Egyptian hieroglyphic texts.

CRediT authorship contribution statement

Duque-Domingo Jaime: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Herrera Pedro Javier:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Fuentes-Ferrer Raúl:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the Agencia Estatal de Investigación under the PID2020-113229RB-C43/AEI/10.13039/501100011033 and PID2020-112658RB-I00/AEI/10.13039/501100011033 Grants. In addition, the research has received funding from the ROSOGAR PID2021-123020OB-I00 project funded by MCIN/ AEI/10.13039/501100011033 / FEDER, UE.

Data availability

Our dataset is publicly accessible on GitHub: https://github.com/rfuentesfe/EgyptianHieroglyphicText.

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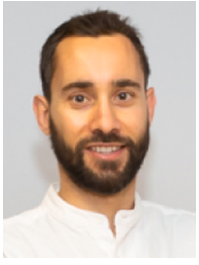
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